# SURJECTIVITY DESCENDS TO A FINITE LEVEL VIA CHEVALLEY'S THEOREM

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*Editorial note.* The step 'The morphism f_0 is of finite presentation' is implicit in the span: locally of finite presentation plus the quasi-compact source of the Situation gives finite presentation; a mechanical one-step fact, recorded here.

*Allowed standard prerequisites (editorial).*

- Chevalley's theorem. Morphisms, Lemma morphisms-lemma-chevalley
- Limit contained in constructible. Limits, Lemma lemma-limit-contained-in-constructible (same chapter)

*Source reference.* Original reference 081D: sources/stacks-chow/tags.txt, source lines 10213–10213. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 054J: sources/stacks-chow/tags.txt, source lines 6642–6642. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original lemma 05F4: sources/stacks-chow/tags.txt, source lines 7013–7013. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Notation and assumptions as in Situation 081D. If*

(1) *$f$ is surjective, and*
(2) *$f_0$ is locally of finite presentation,*

*then there exists an $i \geq 0$ such that $f_i$ is surjective.*

*Proof.* The morphism $f_0$ is of finite presentation. Hence $E = f_0(X_0)$ is a constructible subset of $Y_0$, see Morphisms, Lemma 054J. Since $f_i$ is the base change of $f_0$ by $Y_i \to Y_0$ we see that the image of $f_i$ is the inverse image of $E$ in $Y_i$. Moreover, we know that $Y \to Y_0$ maps into $E$. Hence we win by Lemma 05F4. □

*Date*: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.